

Normalization and the trace identity determine the CAR trace

MathlibAnnex.CStarAlgebra.CAR.eq_trace_of_apply_one_of_mul_comm

theorem

Proves uniqueness without assuming positivity of the competing continuous functional.

Statement

Let A be the completion of $M_{2^n}(\mathbb{C})$ under $a \mapsto a \otimes I_2$, with normalized trace τ . If a continuous complex-linear functional $f : A \rightarrow \mathbb{C}$ satisfies $f(1) = 1$ and $f(ab) = f(ba)$ for every $a, b \in A$, then $f = \tau$.

Assumptions

Continuity, complex linearity, normalization and the trace identity are the hypotheses on f . Positivity, faithfulness and an a priori norm bound of one are not additional hypotheses. Let $\iota_n : M_{2^n}(\mathbb{C}) \rightarrow A$ be the canonical stage embedding. Write $e_{ij} = \iota_n(E_{ij})$ for the matrix units of a stage of size $d = 2^n$.

Conclusion

There is at most one normalized continuous trace on A , and the continuous functional τ defined in the CAR trace construction supplies it. In particular any tracial state equals τ , but the assertion applies to the larger class of functionals specified above.

The argument depends on continuity at the last density step. It makes no uniqueness assertion for arbitrary discontinuous algebraic functionals, or for states without the trace identity.

Proof route

The trace identity and matrix-unit multiplication force all diagonal values $f(e_{ii})$ to agree and all off-diagonal values to vanish. Since $\sum_i e_{ii} = 1$, each diagonal value equals $1/d$. Hence f equals the normalized matrix trace on every stage. The union of stage images is dense in A , so continuity gives equality everywhere.

▼ Proof steps

1. Apply the trace identity to $e_{i0}e_{0i} = e_{ii}$ and $e_{0i}e_{i0} = e_{00}$ to obtain equal diagonal values.
2. For $i \neq j$, the products $e_{ii}e_{ij} = e_{ij}$ and $e_{ij}e_{ii} = 0$ give $f(e_{ij}) = 0$. Normalization now gives $df(e_{00}) = 1$.
3. Expand any stage matrix in matrix units. Linear equality on all stages extends to the completion because f and τ are continuous and the stage union is dense.

Main citations

- Exact statement and proof
- The continuous trace on the completion
- Matrix-unit multiplication
- The diagonal matrix units sum to one
- Expansion in matrix units
- Trace values forced on matrix units
- Uniqueness on each matrix stage
- Density of the finite-stage union
- Normalization at the unit
- The trace identity

Lean source signature (exact)

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theorem eq_trace_of_apply_one_of_mul_comm
  (f : Limit → L[ $\mathbb{C}$ ]  $\mathbb{C}$ ) (hf1 : f 1 = 1)
  (hf :  $\forall a b, f (a * b) = f (b * a)$ ) : f = trace
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In the source	Mathematical meaning
$f : \text{Limit} \rightarrow L[\mathbb{C}] \mathbb{C}$	A given continuous complex-linear functional $f : A \rightarrow \mathbb{C}$, with A the completed CAR algebra.
$\text{hf1} : f 1 = 1$	The normalization $f(1_A) = 1 \in \mathbb{C}$.
$\text{hf} : \forall a b, f (a * b) = f (b * a)$	For every ordered pair $a, b \in A$, assume $f(ab) = f(ba)$. This is the trace identity, with no positivity assumption.
$f = \text{trace}$	The conclusion $f = \tau$: $f(x) = \tau(x)$ for every $x \in A$, where τ is the normalized continuous CAR trace.

Further source notes: Thus the source conclusion $f = \text{trace}$ is the uniqueness assertion in the Statement, without a positivity hypothesis.

Earlier published Card and PDF

▼ Exact content identity

Declaration: MathlibAnnex.CStarAlgebra.CAR.eq_trace_of_apply_one_of_mul_comm

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